

Radio Recombination Line Forecasts with CHORD

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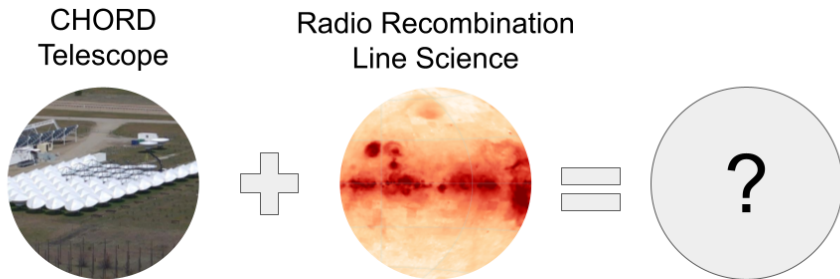
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1

Motivation & Background

- Research Goal
- Physics Background

1.1 Motivation



- Will CHORD detect Hydrogen Radio Recombination Lines?
- If so, what angular resolution and sensitivity can be achieved?

1.2 The Canadian Hydrogen Observatory and Radio-transient Detector (CHORD)

- CHORD is a drift scan interferometer radio telescope:
 - **Drift scan:** the telescope is fixed and the sky drifts overhead.
 - **Interferometer:** 512 dishes work together to form a single sky image.
 - **Radio telescope:** CHORD measures from 300 – 1500 MHz.
- CHORD Pathfinder will be operational in 2026, with the full CHORD array operational in 2028.

CHORD under construction

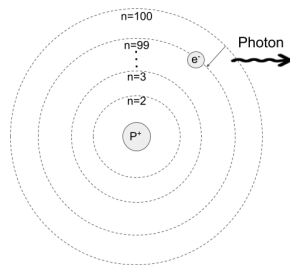


CHORD is situated at the Dominion Radio Astrophysical Observatory (DRAO) (The CHORD Collaboration 2026a)

1.2 Radio Recombination Lines

- When an electron recombines with an ion, it emits a stream of photons.
- Each transition emits a photon of a different frequency.
- Radio recombination lines (RRLs) are photons in the radio range.

Bohr atomic model



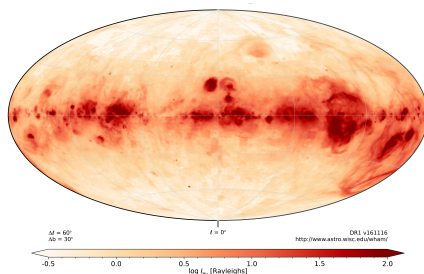
Simplified Bohr model of the hydrogen atom showing RRL emission from $N = 100 \rightarrow 99$.

1.2 The Interstellar Medium (ISM)

Why measure RRLs?

- RRLs are produced in ionized regions of the interstellar medium (ISM).
- Studying the ionized ISM teaches us about star formation and galactic evolution.
- RRLs travel further compared to optical emission, providing an extended view of the ISM.

Wisconsin H α Mapper sky survey



All-sky map of H α emission in the visible spectrum showing the ionized ISM (L. M. Haffner et al. 2010).

2

Radio Frequency Interference

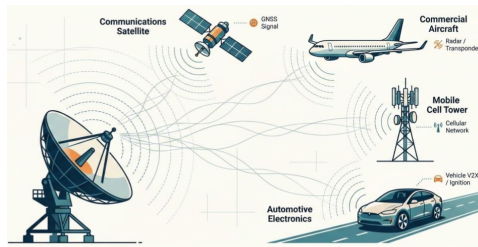
- What is RFI
- How to detect RFI
- RFI impact on Radio Recombination Lines

2.1 Definition of RFI

- Radio Frequency Interference (RFI) are unwanted human-made signals.
- Astronomical Signals are Gaussian distributed, while RFI is non-Gaussian.

To detect an RRL, we need to ensure there is no competing RFI.

Examples of RFI



Sources of RFI can include satellites, aircraft, mobile phones, and cars. (Google 2026)

2.2 RFI-Monitor Data

Analysis of RFI-Monitor data statistics:

1. Complex voltages are measured and converted to power

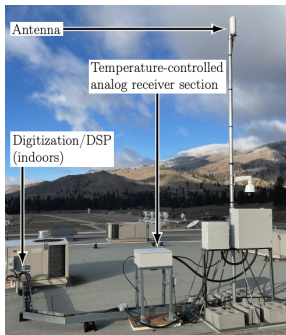
$$P = |V|^2 \sim \text{Exp}(2\sigma^2).$$

2. Power is integrated into $0.75s$ windows over $N = 2500$ time samples

$$P_N = \sum_{i=1}^N P_i \sim \text{Gamma}(N, 2\sigma^2).$$

RFI-monitor data provided by Nick Bruce, DRAO.

DRAO RFI Monitor



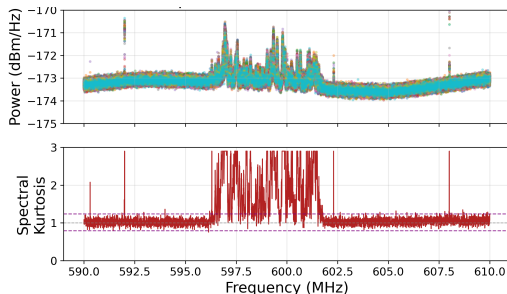
The roof mounted DRAO RFI-Monitor which measures all incoming radio signals (N. Bruce et al. 2026).

2.2 RFI Detection

I applied the generalized Spectral Kurtosis (SK) (G. M. Nita & D. E. Gary 2010) estimator to detect RFI.

- I showed how to calibrate the SK estimator on RFI free data.
- I found that SK must be computed within RFI monitor calibration cycles.
- I showed that the SK estimator could accurately detect RFI.

Power and SK on 600 MHz spectrum



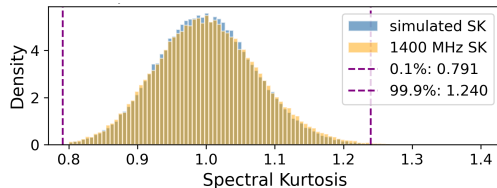
Top plot shows RFI Monitor data. Bottom plot shows calibrated spectral kurtosis.

2.2 Detection Thresholds

I obtained principled RFI detection thresholds and showed they conformed with observed RFI free data.

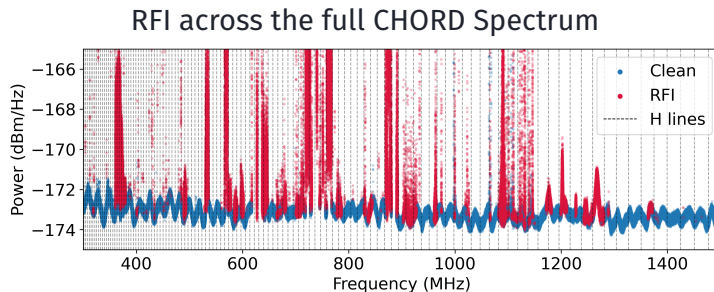
- Simulated the theoretical SK distribution (blue).
- Observed SK distribution (yellow).
- I selected 0.1% and 99.9% percentiles as RFI detection thresholds.

SK distribution of RFI free data



Theoretical SK distribution (blue) matches observed non-RFI SK distribution (yellow).

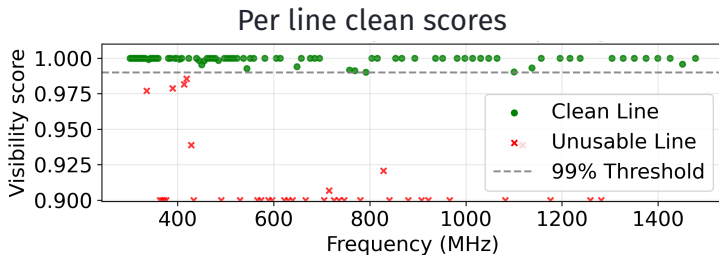
2.3 RFI across the full CHORD Spectrum



RFI-monitor power colored by RFI flagging.

- I found that 75.6% of the CHORD band is RFI free.
- The GNSS band (1215-1300 MHz) shows less RFI than expected.

2.3 Clean Scores



Gaussian weighted average around each RRL with a 100 km s^{-1} standard deviation.

- I found that 79 of the 116 $\text{Hn}\alpha$ RRLs are RFI free.

2.3 RFI Analysis Limitations

- RFI-monitor sensitivity differs from CHORD.
- The RFI-monitor is angled towards the ground and horizon.
- The available data was limited to a single hour.
- The RFI environment might change in the future.

3

CHORD Sensitivity to RRLs

- CHORD sensitivity maps
- Impact of RRL stacking and smoothing
- CHORD sensitivity compared to published SIGGMA survey

3.1 Sensitivity Definitions

Point Source Sensitivity

CHORD's point source sensitivity is computed as

$$\sigma_{\text{RMS}} [\text{Jy/beam}] = \frac{2k (T_{\text{ant}} + T_{\text{sky}})}{A_e \sqrt{2 \Delta\nu \tau_{\text{eff}} N(N-1)}}$$

where

- k is Boltzmann's constant
- T_{ant} is the antenna temperature
- T_{sky} is the sky temperature
- A_e is the effective area of a single dish
- $\Delta\nu$ is the bandwidth
- τ_{eff} is the effective integration time
- N is the number of dishes.

3.1 Sensitivity Definitions

Brightness Temperature Sensitivity

RRLs are extended sources, so we need to convert from point-source sensitivity to brightness temperature sensitivity

$$\sigma_T [\text{K}] = \frac{\sigma_{\text{RMS}}}{\Omega_s} \frac{c^2}{2k\nu^2}$$

where

- Ω_s is the solid angle of the synthesized beam
- c is the speed of light
- ν is the observing frequency

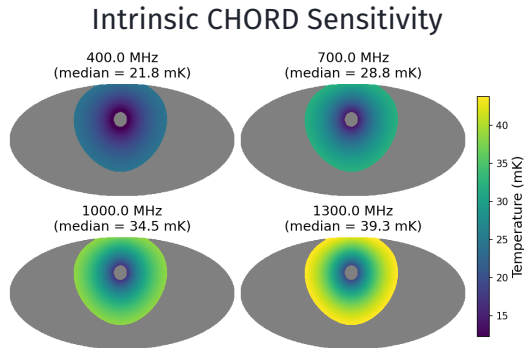
3.1 Sensitivity Maps

Intrinsic CHORD Brightness Temperature Sensitivity

Brightness temperature sensitivity if

$$T_{\text{sky}} = 0$$

- Low Frequency has a larger primary beam.
- High declination has longer integration time.

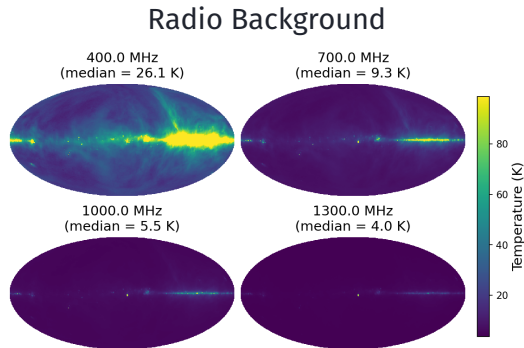


Intrinsic CHORD brightness temperature sensitivity in galactic coordinates.

3.1 Sensitivity Maps

Radio Background

- Radio background is highest at low frequency and along the galactic plane.

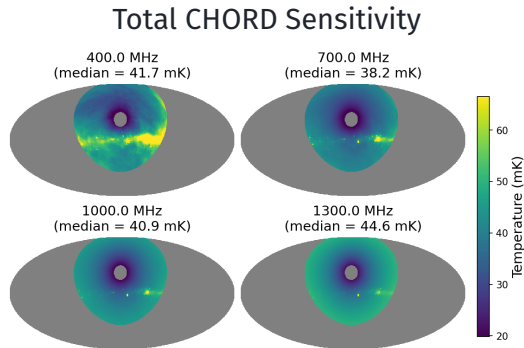


Radio Background obtained from the 2016 Global Sky Model (H. Zheng et al. 2017).

3.1 Sensitivity Maps

Total Sensitivity

- Sensitivity is highly spatially and frequency dependent.



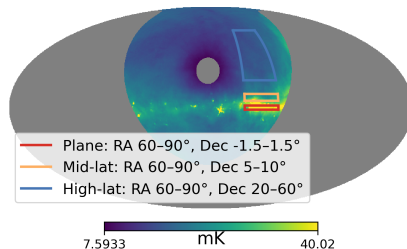
Total CHORD brightness temperature sensitivity over the CHORD observing range.

3.2 Signal to Noise Ratio of RRLs

Selected Regions

What is the signal-to-noise ratio (SNR) of RRL detections in different regions of the sky?

Map of regions for further analysis



We take the median sensitivity over each region, Galactic plane, mid-latitude, and high-latitude.

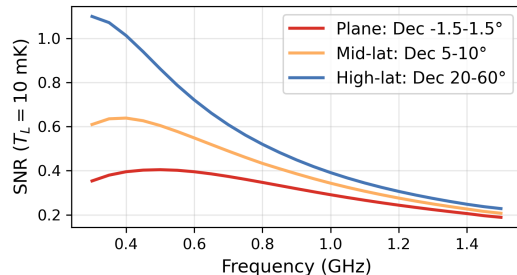
3.2 Signal to Noise Ratio of RRLs

Native SNR per RRL

	Native resolution	
	300 MHz	1.5 GHz
Angular Res.	21'	4.2'
Velocity Res.	5 km s ⁻¹	1 km s ⁻¹

- Low frequency has worse angular and velocity resolution.
- Low frequency has better SNR per line.

SNR by region, using native resolutions



Native SNR per RRL for each region, assuming a line intensity of 10 mK at 1.5 GHz.

3.2 Signal to Noise Ratio of RRLs

SNR with smoothing

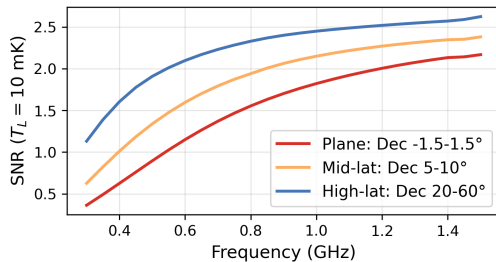
- Each line is smoothed to a 5 km s^{-1} velocity resolution.
- Each line is smoothed to 22 arcminute angular resolution.

$$N = \frac{\Omega_{\text{smoothed}}}{\Omega_{\text{native}}}$$

$$\sigma_T[\text{smoothed}] = \frac{\sigma_T[\text{native}]}{\sqrt{N}}$$

Where N is the independent samples in the smoothed beam and σ_T is the sensitivity.

SNR by region, after smoothing



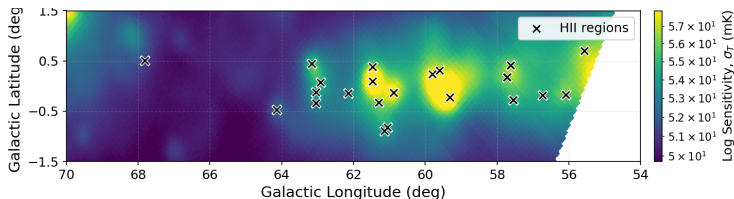
Smoothed SNR shows that high frequency RRLs now have the best SNR per line.

3.3 CHORD detection of SIGGMA Results

SIGGMA range

The Survey of Ionized Gas in the Galaxy Made with the Arecibo telescope (SIGGMA) made 427 RRL detections. 26 are within the CHORD survey area.

RRL detections plotted on CHORD sensitivity map



The CHORD sensitivity at 900 MHz overlaid by the SIGGMA detections.

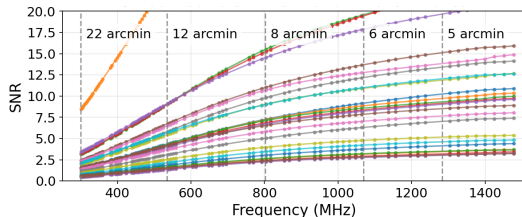
3.3 CHORD detection of SIGGMA Results

SNR per smoothing level

- I used the SIGGMA results to estimate expected Signal.
- I used CHORD sensitivity maps to estimate expected Noise.
- I smoothed to a common angular and velocity resolution.

The number of lines to stack depends on the angular smoothing level.

SNR of SIGGMA detections after smoothing



SNR smoothed to a common 22 arcminute resolution.

3.3 CHORD detection of SIGGMA Results

Detection Results

Stacked SNR per RRL detection with one observing day

Order	Name	SNR 5'	SNR 6'	SNR 8'
1	GO63.164+00.449	85.04 ± 4.97	130.52 ± 7.63	222.38 ± 13.00
2	GO61.473+00.094	41.86 ± 3.23	63.49 ± 4.90	105.47 ± 8.14
⋮	⋮	⋮	⋮	⋮
25	GO62.142-00.144	2.03 ± 0.31	3.13 ± 0.49	5.42 ± 0.84
26	GO57.715+00.173	2.02 ± 0.20	3.11 ± 0.31	5.32 ± 0.53

Smoothing to 5', 6', 8' stacks 8, 14, and 27 lines respectively.

Conclusion: CHORD can detect all SIGGMA results within a single observing day.

3.3 Sensitivity Analysis Limitations

- The radio background used for T_{sky} is limited to 56 arcminute resolution.
- We assumed the smoothed beam is an average over independent native beams, but true native beams have some spatial noise correlations.
- Frequency resolution is dependent on the upsampling hardware and might have limitations.

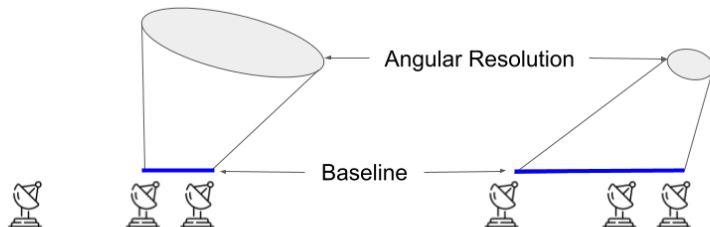
4

Extension CHORD

- How would adding more dishes improve CHORD's angular resolution?

4.1 Why add more dishes?

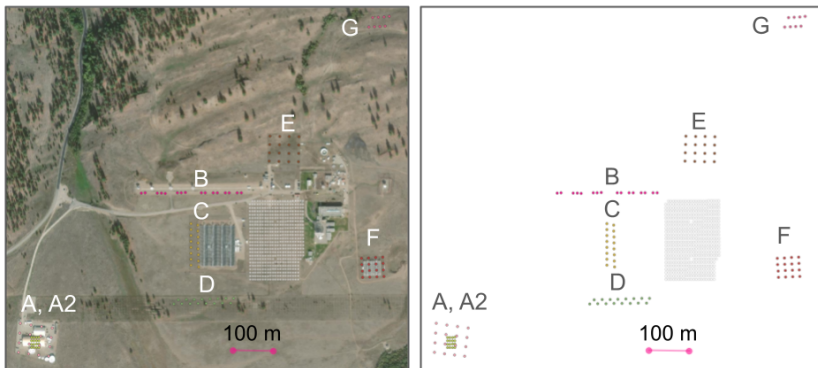
Longer baselines provide smaller angular resolution.



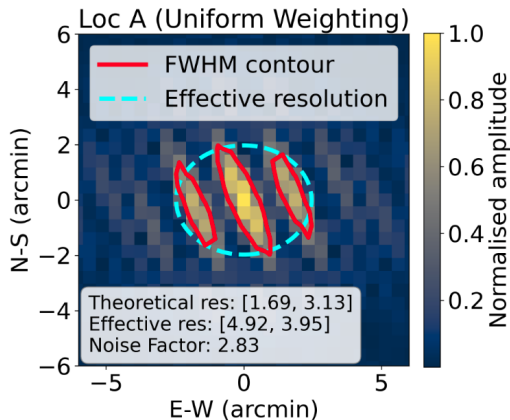
- Would more dishes improve angular resolution to RRLs?
- What is the optimal dish placement?

4.1 Candidate locations

Extension CHORD candidate locations

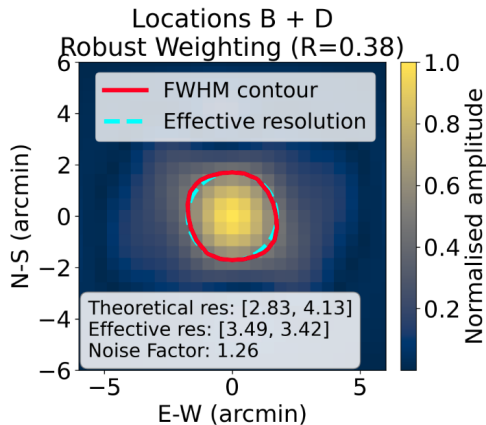


4.1 Location A, the longest possible baseline



Despite having the longest baseline, Location A is not the best option.

4.1 Location B + D, the best overall combination



Angular resolution decreased by 44% while noise increased by 26%.

5 Conclusions

Can CHORD detect RRLs?

- I created an RFI detection pipeline and found 79 of the 116 Hn α RRLs are RFI-free.
- I showed CHORD can reproduce all SIGGMA detections.

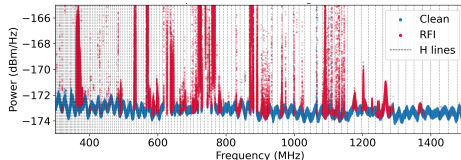
What sensitivity can be achieved?

- I provide sensitivity maps and stacked SNR methods given a desired smoothing.

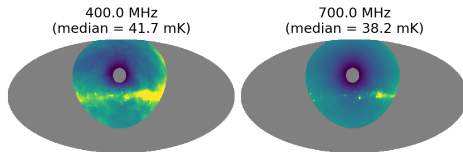
What angular resolution can be achieved?

- Extension-CHORD could yield 44% improvement for a 26% noise cost.

RFI Classifications



Total Sensitivity



6 References I

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- The CHORD Collaboration. 2026a, Overview of the Canadian Hydrogen Observatory and Radio Transient Detector (CHORD) Project, in *SPIE Astronomical Telescopes and Instrumentation*, Vol. Submitted, Copenhagen, Denmark

Zheng, H., Tegmark, M., Dillon, J. S., et al. 2017, An Improved Model of Diffuse Galactic Radio Emission from 10 MHz to 5 THz, Monthly Notices of the Royal Astronomical Society, 464, 3486, doi: 10.1093/mnras/stw2525

7

Appendix



- System noise and astronomical signals are Gaussian distributed over time, while RFI is not.
- Spectral Kurtosis (SK) (G. M. Nita & D. E. Gary 2010) is a measure of Gaussianity in a power timeseries.
- For a Gaussian voltage timeseries, the spectral kurtosis is 1.

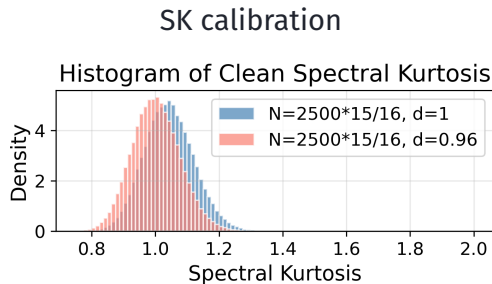
The generalized spectral kurtosis estimator:

$$\text{let: } S_1 = \sum_{j=1}^M P_{N,j}, \quad S_2 = \sum_{j=1}^M P_{N,j}^2$$

$$\widehat{SK} = \frac{MNd + 1}{M - 1} \left(\frac{MS_2}{S_1^2} - 1 \right),$$

Where P_N is the integrated power per frequency channel, M is the number of integrated time samples, Nd is the number of independent time samples per integration.

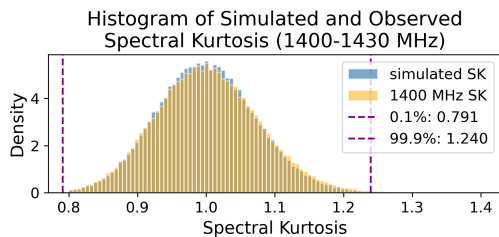
- We used RFI free data from the RFI protected band of 1400-1430 MHz to calibrate the SK estimator.
- This is required since the RFI monitor does not obtain perfectly independent time samples, which biases the estimator.
- The calibrated shape parameter is found to be $Nd = 2500 \cdot 15/16 \times 0.96$



The SK estimator (blue) is calibrated on clean data (red) to have a mode of exactly 1.

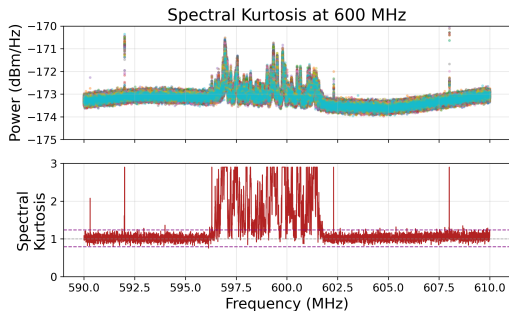
- The RFI-free SK distribution is simulated with 10^5 Monte Carlo draws from $\text{Gamma}(Nd, 2\sigma^2)$.
- The observed SK in the clean 1400–1430 MHz band matches these simulations, confirming the band is RFI free.
- The simulated 0.1% and 99.9% quantiles set the detection thresholds: SK values outside them are flagged as RFI.

Null SK distribution with RFI detection thresholds



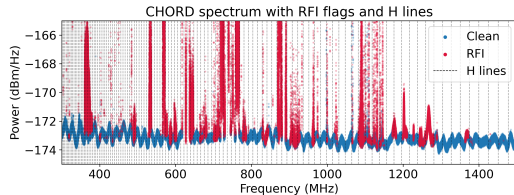
Simulated null SK distribution (blue) matches with observed non-RFI SK (red). RFI thresholds are taken as the 0.1% and 99.9% quantiles.

Power and SK on RFI spectrum



SK (bottom) identifies RFI-affected Power spectrum (top).

RFI flagging results, full CHORD band



RFI monitor power colored by SK RFI flagging results. In total 75.6% of the CHORD band is found to be RFI free.

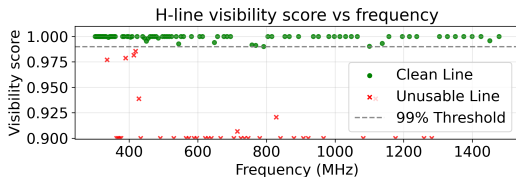
7 Line Characterization

Gaussian weighted average around each RRL

- For a line to be detectable, it should be mostly RFI free within ± 200 km/s.
- We compute a per line score as a Gaussian weighted average of the binary RFI labels centered at the rest frequency with a standard deviation of $\sigma_f = 100$ km/s

$$s = \frac{\sum_i w_i \text{RFI}_i}{\sum_i w_i}, \quad w_i = \exp\left(-\frac{(f_i - f_0)^2}{2\sigma_f^2}\right)$$

Clean Scores



Per line clean scores for all 116 Hn α RRLs in the CHORD band. Lines with a score > 0.99 are considered detectable.

Line Characterization

Detectable Lines by Frequency Range

Detectable vs. RFI-contaminated Hn α RRLs

Frequency Range (MHz)	Detectable Transitions	Contaminated Transitions
300–470	29	10
470–800	23	16
800–1500	27	11
Total	79	37

- CHORD is orders of magnitude more sensitive than the RFI monitor, signals that are not detectable by the RFI monitor may nonetheless impede CHORD.
- CHORD has a directional beam pointed upwards while the RFI monitor is most sensitive downwards and horizontally, RFI from overhead may be missed by the RFI monitor.
- This analysis is based on a single hour of RFI monitor data, which may not be representative of the long-term RFI environment.
- RFI changes over time, as new RFI sources emerge, spectrum currently free of RFI might become corrupted.

A telescope's sensitivity, choosing a signal-to-noise ratio (SNR) of 1, is equivalent to the root mean square (RMS) noise of the telescope.

The point-source RMS noise follows the radiometer equation, which for dual polarization is given by

$$\sigma_{\text{RMS}} [\text{Jy/beam}] = \frac{2k (T_{\text{ant}} + T_{\text{sky}})}{A_e \sqrt{2 \Delta\nu \tau_{\text{eff}} N(N-1)}}$$

where k is Boltzmann's constant, T_{ant} is the antenna temperature, T_{sky} is the sky temperature, A_e is the effective area of a single dish, $\Delta\nu$ is the bandwidth, τ_{eff} is the effective integration time, and N is the number of dishes.

The angular resolution is the smallest angular scale the telescope can resolve. It is defined as the full width at half maximum (FWHM) of the synthesized beam, which is the combined beam after correlating all pairs of dishes in the interferometer.

The angular resolution of an interferometer is set by the longest baseline between two dishes (D_{EW} and D_{NS}) and the observing wavelength (λ),

$$\theta_{FWHM} = \frac{\lambda}{D_{EW}}, \quad \phi_{FWHM} = \frac{\lambda}{D_{NS}}.$$

The beam solid angle is the area subtended by the synthesized beam,

$$\Omega_s = \frac{\pi}{4 \ln 2} \theta_{FWHM} \phi_{FWHM}.$$

Brightness temperature sensitivity, is a measure of the telescope's ability to detect extended sources. It is obtained by dividing the point-source RMS noise by the beam solid angle, and converting from Jy/beam to K by applying the Rayleigh-Jeans limit.

$$\sigma_T [\text{K}] = \frac{\sigma_{\text{RMS}}}{\Omega_s} \frac{c^2}{2k\nu^2}$$

Emission intensity of an RRL is a function of the physical quantities of the ISM and the transition frequency. Peak line strength is given by L. D. Anderson et al. (2021) as

$$\left(\frac{T_L}{\text{K}}\right) \simeq 3076 \left(\frac{T_e}{\text{K}}\right)^{-3/2} \left(\frac{EM}{\text{pc cm}^{-6}}\right) \left(\frac{\nu_0}{\text{GHz}}\right)^{-1} \left(\frac{\Delta V}{\text{km s}^{-1}}\right)^{-1} \Delta n M_{\Delta n},$$

where:

- T_e , the electron temperature, EM , the emission measure, and ΔV , the line width, are physical quantities of the ISM.
- $\Delta n M_{\Delta n}$ is a quantum mechanical factor that is constant for all $n\alpha$ transitions.
- ν_0 is the rest frequency of the transition.
- **Meaning Line strength increases at lower frequencies.**

7 Stacking

- RRLs of different frequencies are all produced by the same underlying physical quantities; gas temperature and density.
- Measurement uncertainty is reduced by averaging signal from multiple RRLs.
- Inverse-variance weighting is used to combine signals having different SNR levels.

Inverse-variance weighted average:

$$\hat{\mu} = \frac{\sum_i^N x_i w_i}{\sum_i^N w_i}, \quad \text{Var}(\hat{\mu}) = \frac{1}{\sum_i^N w_i},$$

$$w_i = \frac{1}{\sigma_i^2},$$

$$\text{SNR} = \frac{\hat{\mu}}{\sqrt{\text{Var}(\hat{\mu})}} = \frac{\sum_i^N x_i w_i}{\sqrt{\sum_i^N w_i}}.$$

Combined signal ($\hat{\mu}$), uncertainty ($\text{Var}(\hat{\mu})$) and SNR after stacking N lines.